

PCA, Mahalanobis Distance, and Outliers

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Contents

1	Simulated Data	1
2	PCA	1
3	A Second Round	5
4	A Final Round	8
5	Appendix	8

1 Simulated Data

We simulate a dataset.

```
> set.seed(564684)
> nSamples <- 30
> nGenes <- 3000
> dataset <- matrix(rnorm(nSamples*nGenes), ncol=nSamples, nrow=nGenes)
> dimnames(dataset) <- list(paste("G", 1:nGenes, sep=''),
+                             paste("S", 1:nSamples, sep=''))
```

Now we make two of the entries into distinct outliers.

```
> nShift <- 300
> affected <- sample(nGenes, nShift)
> dataset[affected,1] <- dataset[affected,1] + rnorm(nShift, 1, 1)
> dataset[affected,2] <- dataset[affected,2] + rnorm(nShift, 1, 1)
```

2 PCA

We start with a principal components analysis (PCA) of this dataset. A plot of the samples against the first two principal components (PCs) shows two very clear outliers (**Figure 1**).

```
> library(ClassDiscovery)
> spca <- SamplePCA(dataset)
```

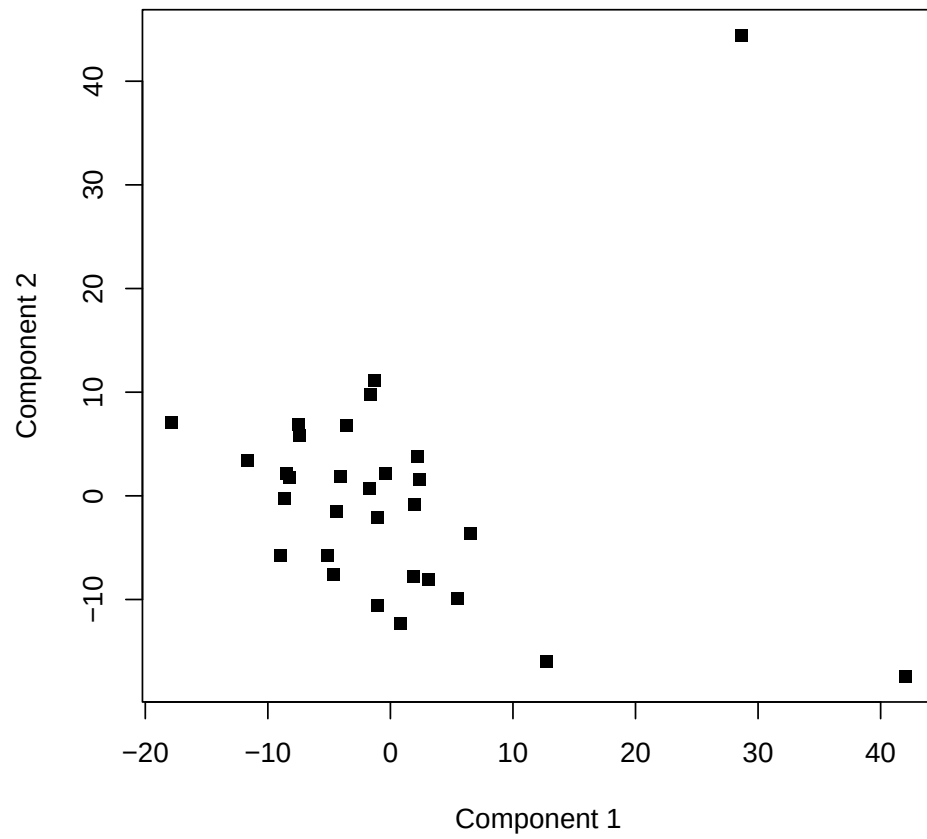


Figure 1: Principal components plot of the samples.

We want to explore the possibility of an outlier more formally. First, we look at the cumulative amount of variance explained by the PCs:

```
> round(cumsum(spca@variances)/sum(spca@variances), digits=2)

[1] 0.04 0.08 0.12 0.16 0.20 0.24 0.28 0.31 0.35 0.39 0.42 0.46 0.49 0.53 0.56 0.60
[17] 0.63 0.66 0.69 0.73 0.76 0.79 0.82 0.85 0.88 0.91 0.94 0.97 1.00 1.00
```

We see that we need 20 components in order to explain 70% of the variation in the data. Next, we compute the Mahalanobis distance of each sample from the center of an N -dimensional principal component space. We apply the `mahalanobisQC` function using different numbers of components between 2 and 20.

```
> maha2 <- mahalanobisQC(spca, 2)
> maha5 <- mahalanobisQC(spca, 5)
> maha10 <- mahalanobisQC(spca, 10)
> maha20 <- mahalanobisQC(spca, 20)
> myd <- data.frame(maha2, maha5, maha10, maha20)
> colnames(myd) <- paste("N", rep(c(2, 5, 10, 20), each=2),
+                          rep(c(".statistic", ".p.value"), 4), sep='')
```

The theory says that, under the null hypothesis that all samples arise from the same multivariate normal distribution, the distance from the center of a d -dimensional PC space should follow a chi-squared distribution with d degrees of freedom. This theory lets us compute p -values associated with the Mahalanobis distances for each sample (**Table 1**). We see that the samples S1 and S2 are outliers, at least when we look at the first 2, 5, or, 10 components. However, sample S2 is not quite significant (at the 5% level) when we get out to 20 components. This can occur when there are multiple outliers because of the “inflated” variance estimates coming from the outliers themselves.

	N2.statistic	N2.p.value	N5.statistic	N5.p.value	N10.statistic	N10.p.value	N20.statistic	N20.p.value
S1	27.4	0.0000	32.6	0.0000	34.1	0.0002	38.3	0.0083
S2	43.1	0.0000	43.4	0.0000	44.7	0.0000	46.8	0.0006
S3	0.2	0.9269	0.7	0.9850	8.7	0.5634	22.5	0.3161
S4	0.4	0.8090	2.0	0.8460	3.4	0.9711	18.0	0.5862
S5	0.5	0.7778	0.6	0.9860	1.9	0.9969	31.3	0.0510
S6	0.5	0.7969	7.2	0.2076	12.6	0.2485	20.4	0.4332
S7	3.0	0.2281	8.7	0.1197	17.9	0.0558	26.5	0.1502
S8	1.0	0.6038	2.5	0.7801	11.7	0.3036	20.0	0.4606
S9	0.6	0.7563	1.5	0.9157	9.7	0.4643	24.7	0.2132
S10	0.0	0.9810	9.1	0.1067	12.9	0.2277	25.3	0.1886
S11	0.5	0.7672	3.1	0.6820	18.6	0.0451	23.6	0.2591
S12	0.9	0.6513	5.7	0.3382	11.7	0.3078	20.5	0.4240
S13	0.9	0.6337	1.1	0.9534	8.5	0.5813	15.8	0.7281
S14	0.0	0.9875	2.5	0.7725	7.0	0.7277	17.2	0.6384
S15	0.1	0.9706	0.2	0.9986	10.1	0.4300	15.2	0.7662
S16	1.2	0.5405	2.7	0.7437	13.3	0.2065	25.8	0.1725
S17	1.0	0.6042	5.7	0.3380	7.6	0.6716	14.7	0.7940
S18	0.6	0.7483	5.3	0.3808	7.5	0.6770	17.2	0.6396
S19	0.5	0.7950	5.1	0.4073	14.9	0.1374	21.7	0.3551
S20	0.8	0.6784	2.1	0.8307	8.3	0.6016	21.0	0.3971
S21	0.8	0.6727	10.9	0.0536	16.9	0.0765	23.8	0.2513
S22	0.6	0.7336	3.5	0.6159	4.3	0.9331	13.0	0.8764
S23	0.2	0.9276	1.9	0.8691	6.2	0.7944	17.9	0.5931
S24	0.2	0.9230	3.8	0.5716	7.2	0.7095	22.9	0.2948
S25	0.6	0.7484	8.7	0.1223	13.8	0.1802	25.4	0.1870
S26	0.0	0.9796	1.7	0.8921	8.9	0.5437	19.3	0.5036
S27	1.1	0.5698	6.0	0.3084	9.4	0.4986	17.0	0.6547
S28	0.0	0.9838	2.8	0.7300	3.6	0.9638	24.8	0.2076
S29	0.7	0.7123	1.9	0.8641	6.4	0.7848	16.7	0.6732
S30	3.4	0.1846	3.5	0.6238	4.9	0.8951	18.4	0.5640

Table 1: Mahalanobis distance (with unadjusted p-values) of each sample from the center of N-dimensional principal component space.

3 A Second Round

Now we repeat the PCA after removing the one definite outlier. Sample S2 still stands out as “not like the others” (**Figure 2**).

```
> reduced <- dataset[,-1]
> dim(reduced)

[1] 3000    29

> spca <- SamplePCA(reduced)
> round(cumsum(spca@variances)/sum(spca@variances), digits=2)

[1] 0.04 0.09 0.13 0.17 0.21 0.24 0.28 0.32 0.36 0.40 0.43 0.47 0.51 0.54 0.58 0.61
[17] 0.65 0.68 0.71 0.75 0.78 0.81 0.85 0.88 0.91 0.94 0.97 1.00 1.00
```

And we can recompute the mahalanobis distances (**Table 2**). Here we see that even out at the level of 20 components, this sample remains an outlier.

```
> maha20 <- mahalanobisQC(spca, 20)
```

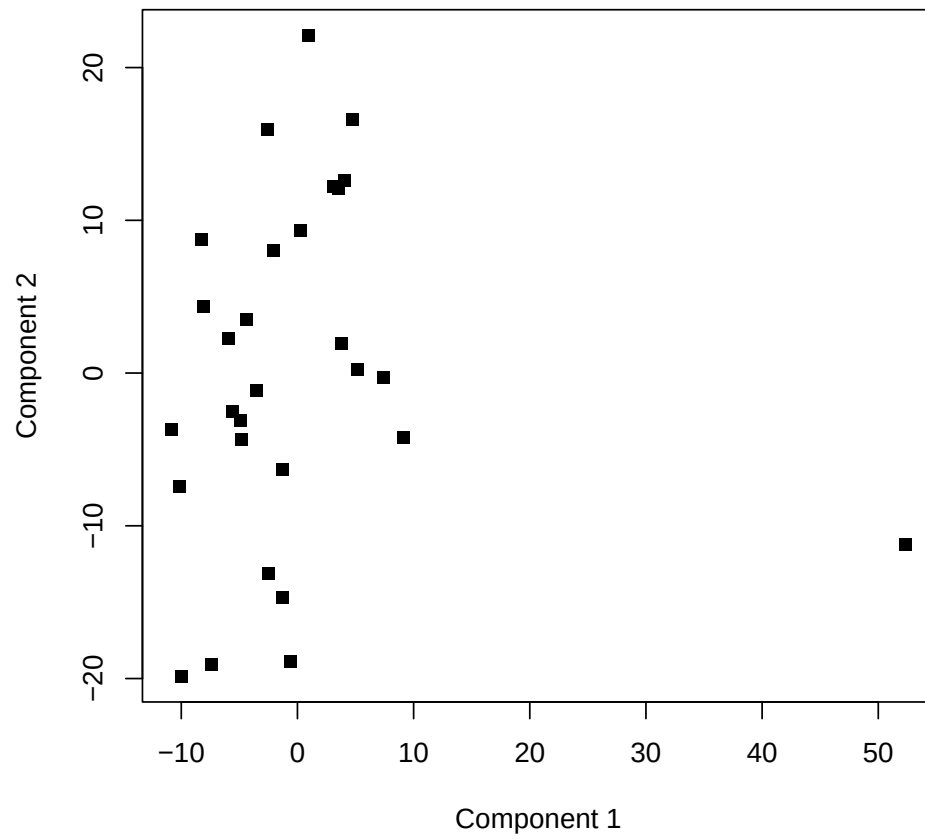


Figure 2: Principal components plot of the normal control samples, after omitting an extreme outlier.

	statistic	p.value
S2	94.8	0.0000
S3	23.5	0.2670
S4	21.5	0.3689
S5	32.6	0.0375
S6	20.6	0.4185
S7	24.7	0.2130
S8	19.5	0.4901
S9	24.8	0.2099
S10	23.9	0.2461
S11	20.9	0.4019
S12	19.3	0.5029
S13	15.3	0.7603
S14	17.4	0.6271
S15	30.2	0.0673
S16	25.8	0.1712
S17	18.8	0.5357
S18	17.2	0.6378
S19	22.4	0.3214
S20	20.3	0.4363
S21	23.6	0.2596
S22	11.7	0.9246
S23	17.5	0.6188
S24	22.0	0.3430
S25	24.3	0.2277
S26	18.3	0.5677
S27	16.6	0.6769
S28	28.1	0.1079
S29	16.3	0.6952
S30	22.7	0.3039

Table 2: Mahalanobis distance (with unadjusted p-values) of each sample from the center of 20-dimensional principal component space.

4 A Final Round

We repeat the analysis after removing one more outlier.

```
> red2 <- reduced[,-1]
> dim(red2)

[1] 3000    28

> spca <- SamplePCA(red2)
> round(cumsum(spca@variances)/sum(spca@variances), digits=2)

[1] 0.04 0.09 0.13 0.17 0.21 0.25 0.29 0.33 0.37 0.41 0.45 0.48 0.52 0.56 0.59 0.63
[17] 0.67 0.70 0.74 0.77 0.81 0.84 0.87 0.91 0.94 0.97 1.00 1.00
```

And we can recompute the mahalanobis distances (**Table 3**). At this point, there are no outliers.

```
> maha20 <- mahalanobisQC(spca, 20)
```

5 Appendix

This analysis was performed in the following directory:

```
> getwd()

[1] "C:/Users/KRC/AppData/Local/Temp/RtmpsJY2Ij/Rbuild34a41b04498f/ClassDiscovery/vignettes"
```

This analysis was performed in the following software environment:

```
> sessionInfo()

R version 4.5.0 (2025-04-11 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26100)

Matrix products: default
  LAPACK version 3.12.1

locale:
[1] LC_COLLATE=C                      LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8 LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

time zone: America/New_York
tzcode source: internal

attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods   base

other attached packages:
[1] xtable_1.8-4      ClassDiscovery_3.4.8 oompaBase_3.2.10    cluster_2.1.8.1

loaded via a namespace (and not attached):
[1] compiler_4.5.0  mclust_6.1.1      oompaData_3.1.5    tools_4.5.0
```

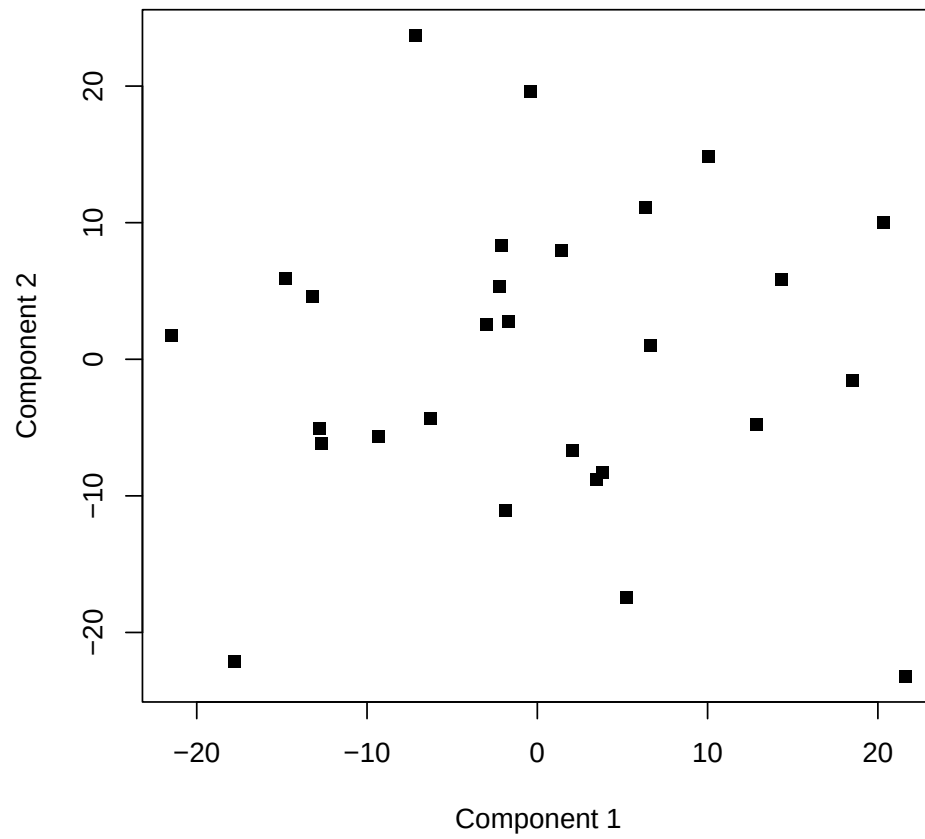



Figure 3: Principal components plot of the normal control samples, after omitting an extreme outlier.

	statistic	p.value
S3	25.3	0.1896
S4	21.3	0.3773
S5	22.9	0.2913
S6	20.0	0.4575
S7	23.8	0.2533
S8	19.7	0.4789
S9	23.5	0.2629
S10	23.4	0.2679
S11	21.5	0.3681
S12	19.3	0.5043
S13	18.4	0.5580
S14	17.2	0.6406
S15	26.4	0.1544
S16	26.4	0.1528
S17	19.4	0.4967
S18	16.8	0.6671
S19	21.8	0.3506
S20	20.2	0.4430
S21	23.0	0.2878
S22	28.4	0.0993
S23	17.0	0.6512
S24	21.5	0.3703
S25	23.4	0.2696
S26	17.6	0.6120
S27	16.7	0.6732
S28	27.8	0.1151
S29	16.3	0.6983
S30	22.4	0.3201

Table 3: Mahalanobis distance (with unadjusted p-values) of each sample from the center of 20-dimensional principal component space.